

YITONG (TIFFANY) LI

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EXPERIENCE

Roblox, Engine UI [C++, Lua] San Mateo, CA | April 2022 - Present

Senior Software Engineer

- Implemented proprietary styling API similar to CSS across engine, application, and plugin stacks, driving its internal and external adoption that resulted in a measured 30% UI code reduction and 23% UI performance improvement
- Owned a project to extend UI corner-rounding capabilities end-to-end, including technical investigation, task planning and delegation, cross-team and cross-discipline alignment, fullstack implementation, and external communications
- Instrumented telemetry end-to-end, sending events from engine to Kafka/Hive, schematizing them, and ingesting them into long-term retention tables to create Superset dashboards that report feature usage, adoption, and performance
- Coordinated resolution of live incidents by advising release engineers, investigating breakages, communicating with affected users, and writing retrospectives that identify process and testing gaps and propose actions to address them
- Regularly resolved community-reported UI bugs, including communicating directly with affected users
- Routinely monitored performance and crash metrics and investigated observed anomalies, including resolving an asset delivery slowdown that resulted in a profiled 13.5 ms reduction in rendering paths for a top-25 game

Riot Games, Valorant [C++, Unreal 4] Los Angeles, CA | August 2020 - March 2022, Summer 2019

Software Engineer

- Supported character KAY/O's release by implementing designer-friendly systems for prototype mechanics, such as temporary death and ability suppression, and conducting blueprint reviews for game designers
- Designed and implemented cached and roaming agent-specific keybind settings through UMG and Unreal engine modifications, with iteration support from QA, designers, and internal playtesters
- Resolved gameplay and networking bugs on two-week patch cycles by reporting bugs to Jira, implementing bugfixes, and coordinating validation and deployment plans with QA

Carnegie Mellon, Robotics Institute [Emscripten, Apache, Shibboleth] Pittsburgh, PA | Summer 2020

Research Intern

- Developed web framework for generating interactive gameplay engineering C++ lessons from doxygen-based source files, serving WebAssembly compiled with Emscripten from student C++ submissions, and delivering instructor feedback on submissions
- Deployed website to production Apache server with SSL certification and single-service sign-on using Shibboleth

Carnegie Mellon School of Computer Science, Principles of Software Construction [Java] Pittsburgh, PA | Fall 2019

Teaching Assistant

Raven Software, Call of Duty Cold War [C++] Middleton, WI | Summer 2018

Gameplay Engineering Intern

- Iterated on multiplayer game modes by designing, scripting, and group-playtesting levels using proprietary tools

PROJECTS

Magpie, Computer Game [C++, OpenGL, Maya, Blender] Pittsburgh, PA | Fall 2018

- Modeled and animated all 3D assets in Maya and Blender, including characters, environment, and interactables
- Designed and implemented level creator and dynamic asset loading pipeline by encoding mesh, room, and AI pathing information in RGB channels of PNG files

Pick Me Up, Animation Short [Maya, Premiere] Pittsburgh, PA | Spring 2018

- Pitched animatics and character designs through two rounds of selection to recruit and direct a team of 5 animators
- Constructed body rigs for squirrel and turtle characters using dynamic ribbon splines and integrated FK/IK support
- Animated and rendered 13 shots by blocking elements, framing camera, and animating body movements

EDUCATION

Carnegie Mellon University Pittsburgh, PA | May 2020

QPA 3.77/4.0 Bachelor of Computer Science and Arts & Game Design Minor

(Dean's List Fall '16 - Spring '17, Spring '18 - Spring '19)

Courses Computer Game Programming, Computer Graphics, Introduction to Computer Systems, Discrete Differential Geometry, Web App Development, Principles of Software Construction, Character Rigging, RPG Writing Workshop

SKILLS

Programming Languages C++, Lua, Java, C#, Python, HTML5/CSS, JavaScript, SML

Tools Git, Jira, Confluence, Unreal 4, Perforce, Maya, Unity, Blender, Photoshop, Premiere